

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Staff Associate Technical Intern”

Submitted in partial fulfillment for the award of degree (20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Namitha K - 1DS20AI032

Conducted at



American Telephone and Telegraph

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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Department of Artificial Intelligence and Machine Learning



CERTIFICATE

This is to certify that the Internship titled “**Salesforce Developer**” carried out by **Ms Namitha K [1DS20AI032]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect to the course Internship (20AI8ICINT)

**Signature of Reporting
Head**
**Subramaniam
Swamynathan**
Team Lead
AT&T

**Signature of Internship
Coordinator**
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi

Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Namitha K**, final year student of Artificial Intelligence and Machine Learning Engineering, Dayananda Sagar College Of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **AT&T**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:

Place: Bangalore

USN: 1DS20AI032

NAME: NAMITHA K

OFFER LETTER



**AT&T Communication
Services India Pvt. Ltd.**
13th Floor, Mohan Dev
House 13 Tolstoy Marg,
New Delhi 110001, India

Tel: 91. 11.42408774
Fax: 91. 11.42408775
91. 11.42408774
www.ap.att.com

CIN: U64203DL1996PTO78375

December 5, 2023

To,

Namitha K
Dayananda Sagar College of Engineering

REF: INTERNSHIP AND PRE-PLACEMENT OFFER WITH AT&T COMMUNICATION SERVICES INDIA PVT LTD.

Dear Namitha,

With reference to the interview that you have had with us, it gives us great pleasure to offer you an **Internship at AT&T Communication Services India Pvt Ltd.** (the "Company"). In addition to confirming the offer, this letter will describe the terms and conditions of your Pre-placement offer. We hope that this opportunity will provide you with exposure to technology and corporate work culture besides helping you to fulfill the criteria for your academic course.

Title: Your title will be '**Staff Associate Technical Intern**'. You will be informed about your manager by the Company at the time of joining and subsequently whenever there is a change in manager.

Terms and Conditions:

Location: You will report for joining at **Bangalore** office. Post onboarding training you will be required to work from Hyderabad or Chennai office as per business requirement.

Effective Date: Your internship duration will be **Six Months** from **February 1, 2024** or other mutually agreed date and may be extended by mutual consent based on performance appraisals and progress of learning.

Timings: The work hours during the internship will be 8 hours per day besides a meal break of one hour as per the company policy, which may include shift timings.

Payment Terms: This internship is for the purpose of training only and will not be deemed to be employment. You will receive a stipend of **Rs. 45,356.00** per month during your period of internship, less all applicable taxes. This stipend is assuming attendance on all the company working days during the month; stipend will be paid pro-rata in the event of absence on any grounds. Payment of stipend will be as per the Company's prevailing practice, in arrears, for the completed month.

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Job Confirmation: On successful completion of 3 months from the start date, your performance will be evaluated and based on the feedback you may be eligible for a permanent position with AT&T subject to the terms and conditions of the company. You will receive a confirmation letter from the Company, if selected for the permanent position. Your internship will be continued till the end of your academic's course.

Leave: Intern will be eligible for one day of casual/sick leave per month and any leave/s consumed beyond the eligibility will be treated either as Loss of Pay (LOP) or will be added to the internship duration as applicable.

Salary for Permanent Position: You will be eligible for an Annual Compensation in the range of **minimum INR 12,00,000/Annum to INR 16,00,000/Annum** (CTC - Cost to Company) depending on the role of your final placement, as a regular employee of the Company.

Student Verification: The internship offer is subject to your providing a letter from your college/training institution certifying your registration as a student at the college/training institution for the duration of the internship. Student internship will be terminated if the intern is no longer a student in the college/training institute.

Confidentiality, Non-Disclosure, Network & Application Use and Intellectual Property Rights: You will be required to adhere to the terms given in Annexure I to this offer letter. The Annexure mentioned shall be deemed to be integral part of this offer document.

Conduct: You will abide by the rules and regulations of the Company as may be enforced from time to time, including dress code and ethical conduct as applicable for other personnel in the Company.

At-Will Contract: Your Internship contract with the Company is "At-Will." This means that you have the right to terminate your Internship contract at any time and for any reason. Likewise, the Company may terminate your Internship contract with or without cause at any time and for any reason. Accordingly, this letter is not to be construed or interpreted as containing any guarantee of continuing/assured internship. As such, the recitation of certain time periods in this letter is also not to be construed or interpreted as containing any guarantee of internship period or any particular level or nature of compensation.

Background Verification: Your appointment for Internship will be subject to successful completion of background verification.

Indemnity: Your Internship contract is conditional to your providing an Indemnity Undertaking to the Company as Annexure II to this offer letter. Failure to provide this Indemnity Undertaking will render your Internship contract null and void.



This letter (together with the attached Annexures) reflects the entire agreement regarding the terms and conditions of your internship contract. Accordingly, it supersedes and completely replaces any prior oral or written communication on this subject. This letter may not be modified or amended except by a written agreement, signed by the Company and by you.

We welcome you onboard and expect that our relationship will be mutually rewarding. As a token of your acceptance, please sign one copy of the offer letter and return it to the Company.

Sincerely,

For AT&T Communication Services India Pvt. Ltd.

Mohan N Digitally signed by
Mohan N
Date: 2023.12.07
09:28:38 +05'30'

Authorized Signatory

Date:

Acknowledged and Agreed

Namitha K

Candidate Name: Namitha K

Date: 2023. 12. 07

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COMPLETION CERTIFICATE

A C K N O W L E D G E M E N T

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out the Internship and for her valuable guidance and support.

We would like to thank Mr Subramaniam Swamynathan, AT&T , for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, Prof. Kavya D N, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of the AI & ML department for the support extended during the course of the Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not least, we would like to thank our parents and friends without whose constant help, the completion of the Internship would have not been possible.

NAMITHA K [1DS20AI03]

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CHAPTER 1

INTRODUCTION

MuleSoft stands at the forefront of modern integration technology, offering a robust platform for designing, deploying, and managing APIs tailored to specific business requirements. As a Staff Associate Technical Intern specializing in MuleSoft at ABT, the role involves monitoring APIs to ensure they meet the company's standards while continually enhancing functionality. Since the commencement of the internship in February 2024, immersion in the dynamic world of MuleSoft has been paramount, where innovation and efficiency converge to drive seamless connectivity solutions.

At the heart of MuleSoft's appeal is its unique architecture, characterized by three distinct layers: Experience, Process, and System. This innovative framework facilitates API-led connectivity, enabling businesses to effortlessly interconnect various layers and systems without the need for extensive redesign. The concept of API reusability is central to MuleSoft's approach, fostering agility and scalability across diverse business models.

Design and development endeavors take place within the intuitive simulation tool, Anypoint Studio. Here, meticulous crafting of API designs occurs, adding essential fields and parameters before seamlessly integrating them into the Anypoint platform. This streamlined process ensures that updates and modifications are propagated globally, empowering organizations to adapt swiftly to evolving market demands and technological landscapes. In the realm of digital transformation, MuleSoft remains a trusted ally, empowering businesses to unlock the full potential of their data and systems through seamless integration and connectivity.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

AT&T Inc., originally the American Telephone and Telegraph Company, stands as a prominent American multinational telecommunications conglomerate, with its headquarters situated at the imposing Whitacre Tower nestled in Downtown Dallas, Texas. Renowned for its expansive reach and significant revenue, it proudly holds the title of the world's fourth-largest telecommunications entity, a testament to its enduring influence in the industry. Particularly noteworthy is its status as the largest wireless carrier within the United States, a position that underscores its pivotal role in shaping the nation's telecommunications landscape.

Delving into its rich history unveils a narrative steeped in innovation and evolution. For much of the 20th century, AT&T held a monopolistic grip on phone service across the United States, a testament to its unrivaled dominance within the telecommunications sphere. Its origins trace back to the modest beginnings of the American District Telegraph Company, established in St. Louis in 1878. Through a strategic series of mergers and expansions into territories such as Arkansas, Kansas, Oklahoma, and Texas, the company metamorphosed into the Southwestern Bell Telephone Company by 1920. This entity operated as a subsidiary under the umbrella of the American Telephone and Telegraph Company, itself a descendant of the pioneering Bell Telephone Company, founded by none other than Alexander Graham Bell in 1877.

AT&T, with its expansive array of products and services, stands as a comprehensive provider in the telecommunications landscape, catering to diverse consumer needs with efficiency and innovation. From mobile telecom services backed by cutting-edge millimeter wave 5G technology to broadband offerings emphasizing flexibility and choice, AT&T ensures seamless integration of internet services into users' daily lives. Additionally, its satellite TV services and acquisition of Warner Media Group further enhance its entertainment offerings, solidifying AT&T's position as a leader in connectivity and content delivery.

CHAPTER 3

SCOPE OF WORK

The scope of work involves leveraging MuleSoft's integration platform to streamline connectivity and data exchange within the organization. As a Staff Associate Technical Intern, the primary objective is to contribute to the design, deployment, and management of APIs using MuleSoft technologies.

1. **API Monitoring and Compliance:** Monitoring existing APIs to ensure adherence to company standards and specifications is a key responsibility. Regular assessments are conducted to identify areas for optimization and improvement. Collaboration with cross-functional teams is essential to address any compliance issues and implement necessary updates or modifications.
2. **Feature Enhancement and Deployment:** The role includes working on adding new features and functionalities to existing API designs based on business requirements. Collaboration with development teams is crucial to implement new features within the MuleSoft environment. Coordinating with stakeholders ensures smooth deployment of updated APIs in the required environments, including testing and validation processes.
3. **Anypoint Studio Development:** Utilizing Anypoint Studio, the simulation tool provided by MuleSoft, is integral to the role for designing and developing API solutions. Implementing best practices for API design and development within Anypoint Studio ensures scalability, efficiency, and maintainability. Thorough testing and validation of API designs within Anypoint Studio are conducted before deployment to production environments.
4. **Documentation and Reporting:** Maintaining detailed documentation of API designs, configurations, and deployment processes is a vital aspect of the role. Generating reports on API performance, usage metrics, and compliance status facilitates stakeholder and management review. Providing regular updates and progress reports on API development and deployment activities ensures transparency and accountability.

5. Training and Support: Offering training and support to internal teams on MuleSoft integration techniques and best practices is part of the responsibilities. Providing technical assistance and troubleshooting guidance to resolve any issues related to API integration and usage ensures smooth operations. Staying updated on the latest MuleSoft features, enhancements, and industry trends enables continuous improvement in integration processes.

6. Project Management and Collaboration: Collaborating with project managers and cross-functional teams ensures alignment with project timelines and objectives. Participation in regular meetings, workshops, and reviews facilitates discussion on project status, challenges, and opportunities for improvement. Effective communication of project updates, milestones, and deliverables to stakeholders and team members is essential for project success.

7. Quality Assurance and Continuous Improvement: Implementing quality assurance measures to ensure the reliability, scalability, and security of API solutions is a critical aspect of the role. Proactively identifying opportunities for process optimization, automation, and efficiency gains within the MuleSoft integration framework is essential. Participation in post-implementation reviews and lessons learned sessions enables identification of areas for improvement and implementation of corrective actions.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

Learning objectives in a MuleSoft Developer Virtual Internship Program, as outlined by AT&T, may include:

1. Deep Understanding of MuleSoft Integration:

Gain comprehensive knowledge of MuleSoft's integration platform, including its architecture, features, and capabilities, to effectively design, deploy, and manage APIs. 2. Apex Programming: Learn the Apex programming language, which is used for developing custom business logic and integrations within the Salesforce platform.

2. Hands-on Experience with Anypoint Studio:

Acquire proficiency in utilizing Anypoint Studio, MuleSoft's simulation tool, for designing and developing API solutions, ensuring practical experience in API design and development.

3. API Monitoring and Compliance Skills:

Develop skills in monitoring APIs to ensure compliance with company standards and specifications, including the ability to assess and optimize existing APIs for improved performance and functionality.

4. Deployment and Environment Management:

Learn the process of deploying APIs in different environments, including testing, validation, and coordination with stakeholders, to ensure smooth deployment and integration into production environments. 6. Salesforce Automation: Learn how to automate business processes using Salesforce automation tools such as workflows, process builder, and flow builder to streamline operations and improve efficiency.

5. Documentation and Reporting Proficiency:

Enhance documentation and reporting skills to maintain detailed records of API designs, configurations, and deployment processes, as well as generate reports on API performance and compliance for stakeholder review.

6. Collaboration and Communication Abilities:

Improve collaboration and communication skills by working with cross-functional teams, project managers, and stakeholders to align on project objectives, timelines, and deliverables, fostering effective communication channels throughout the project lifecycle.

7. Quality Assurance and Continuous Improvement Mindset:

Cultivate a mindset of continuous improvement by implementing quality assurance measures to ensure the reliability, scalability, and security of API solutions, and proactively identifying opportunities for process optimization and automation within the MuleSoft integration framework.

CHAPTER 5

CHALLENGES AND SOLUTIONS

Navigating through the internship at AT&T, several challenges have emerged, offering valuable opportunities for growth and learning. One notable challenge has been the complexity of understanding and effectively implementing MuleSoft's integration platform. Given its multifaceted architecture and diverse functionalities, grasping the intricacies of Anypoint Studio and API development has presented a significant learning curve.

To address this challenge, proactive measures have been taken to enhance learning and skill acquisition. Engaging in comprehensive training sessions, online tutorials, and hands-on practice with Anypoint Studio has proven instrumental in deepening understanding and proficiency. Additionally, seeking guidance and mentorship from experienced colleagues within the team has provided invaluable insights and support in overcoming technical hurdles and refining integration techniques.

Another challenge has been ensuring API compliance and adherence to company standards while implementing new features and enhancements. Balancing the need for innovation with the imperative of maintaining consistency and reliability in API design and deployment has required careful attention to detail and thorough testing procedures.

To tackle this challenge, a systematic approach to API monitoring and quality assurance has been adopted. Implementing rigorous testing protocols, including unit testing, integration testing, and end-to-end testing, has enabled the identification of potential issues early in the development process. Collaborating closely with stakeholders and subject matter experts has facilitated alignment on compliance requirements and ensured that new features are seamlessly integrated into existing systems without compromising performance or stability.

Furthermore, effective communication and collaboration have been pivotal in overcoming challenges and driving progress. Regular meetings, brainstorming sessions, and knowledge-sharing forums have fostered an environment of open dialogue and collaboration, enabling the team to collectively address challenges and leverage collective expertise to find innovative solutions.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

During my internship at AT&T, I've achieved significant milestones and made notable contributions to the company's API integration initiatives. One of the key accomplishments has been obtaining the MuleSoft Certified Developer (MCD) Level 1 certification, demonstrating proficiency in MuleSoft's integration platform and validating my expertise in API design, development, and management.

In addition to obtaining certification, I've played a pivotal role in various API-related projects, contributing to API monitoring, design, and management. Continuously engaging in API monitoring activities, I've ensured compliance with company standards and specifications, identifying areas for optimization and improvement to enhance overall performance and functionality.

Moreover, my contributions to API design and management have been instrumental in driving project success. By actively participating in user story development and collaborating with cross-functional teams, I've contributed to the seamless integration of new features and enhancements into existing systems, ensuring alignment with business requirements and objectives.

Overall, my achievements and contributions reflect a commitment to excellence and a dedication to driving innovation and efficiency within AT&T's API integration efforts. Moving forward, I remain committed to leveraging my expertise and skills to further advance the company's integration initiatives and contribute to its continued success.

CHAPTER 7

REFLECTION AND ANALYSIS

Reflecting on my internship experience at AT&T, I am struck by the profound impact it has had on my professional growth and development. Immersed in the dynamic world of API integration and MuleSoft technology, I've gained invaluable insights into the complexities of modern telecommunications systems and the critical role of seamless connectivity in driving organizational success. The hands-on experience with Anypoint Studio and the MuleSoft platform has deepened my technical expertise and provided a solid foundation for future endeavors in the field of integration technology.

Moreover, the opportunity to contribute to API monitoring, design, and management projects has been immensely rewarding. Through collaborative efforts with cross-functional teams, I've learned the importance of effective communication, teamwork, and adaptability in navigating complex integration challenges and driving project success. These experiences have not only honed my technical skills but also cultivated a mindset of continuous improvement and innovation, laying the groundwork for future professional endeavors.

Looking ahead, I am excited to continue building upon the knowledge and skills acquired during my internship at AT&T. Armed with a deeper understanding of API integration and MuleSoft technology, along with a newfound appreciation for the importance of collaboration and adaptability in the workplace, I am confident in my ability to tackle future challenges and contribute meaningfully to the advancement of integration initiatives within the organization and beyond.

CHAPTER 8

RECOMMENDATION

Based on my internship experience at AT&T, I wholeheartedly recommend the company as an excellent environment for aspiring professionals seeking to delve into the realm of telecommunications and integration technology. Throughout my time at AT&T, I've been impressed by the company's commitment to fostering a culture of innovation, collaboration, and continuous learning. The opportunities for hands-on experience with cutting-edge technologies like MuleSoft and Anypoint Studio have been invaluable, providing a solid foundation for future career growth in the field of API integration.

Furthermore, the support and mentorship provided by experienced colleagues and supervisors have been instrumental in my professional development journey. Their guidance and encouragement have not only helped me navigate through complex integration challenges but have also inspired me to strive for excellence and embrace new opportunities for growth and learning. The emphasis on teamwork and collaboration at AT&T has fostered a supportive and inclusive work environment where individuals are empowered to leverage their unique strengths and contribute meaningfully to the company's success.

In conclusion, I am confident that any individual embarking on an internship or career journey at AT&T will find themselves surrounded by a community of dedicated professionals committed to driving innovation and excellence in the telecommunications industry. With its focus on continuous learning, collaboration, and mentorship, AT&T provides a conducive environment for aspiring professionals to thrive and make significant contributions to the company's mission and objectives.

CHAPTER 9

CONCLUSION

In conclusion, my internship experience at AT&T has been nothing short of transformative. Through hands-on experience with MuleSoft technology, collaborative projects, and mentorship from seasoned professionals, I've gained invaluable insights into the intricacies of API integration and telecommunications. The challenges I've faced have served as opportunities for growth, strengthening my technical skills, fostering teamwork and communication, and instilling a mindset of continuous improvement.

As I reflect on my time at AT&T, I am grateful for the knowledge, experiences, and relationships that I've gained. Moving forward, I am excited to apply what I've learned to future endeavors, confident in my ability to navigate complex integration challenges and contribute meaningfully to the advancement of technology and innovation. I am optimistic about the possibilities that lie ahead and look forward to leveraging my internship experience at AT&T as a springboard for future success in the dynamic world of telecommunications and integration technology.

CHAPTER 10

REFERENCES

- [1] <https://en.wikipedia.org/wiki/AT%26T>
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- [4] <https://blogs.mulesoft.com/dev-guides/augment-api-elements-from-event-driven-architecture/>